

EE 474 Introduction to Control Systems**Homework 1**

Due Wednesday September 16, 2026

Announcements :

- Attend your laboratory section starting with the second week of class.

Matlab Resources :

If you are not familiar with Matlab, you will need to learn to use Matlab as a self-study activity. Some materials to assist you include:

- Well produced training materials, including short video lessons

http://www.mathworks.com/academia/student_center

- A guide to additional Matlab learning materials can be found at:

http://www.cs.ubc.ca/~murphyk/Software/matlab_tips.html

Reading:

- Franklin, Powell and Emami-Naeini: Preface and Chapter 1.
- Lecture notes, Part 1.

Assignment :

1 Text problems¹ :

- A) Text problem 1.1. Carry out the exercise for two of the listed systems. You select the two systems.

Text problem 1.1 : Draw a component block diagram for each of the following feedback control systems.

- (a) The manual steering system of an automobile
- (b) Drebbel's incubator
- (c) The water level controlled by a float and valve
- (d) Watt's steam engine with fly-ball governor

In each case, indicate the location of the elements listed below and give the units associated with each signal.

- The process
- The process desired output signal
- The sensor
- The actuator
- The actuator output signal
- The controller
- The controller output signal
- The reference signal
- The error signal

Notice that in a number of cases the same physical device may perform more than one of these functions.

- B) Text Problem 1.2

Text problem 1.2 : Identify the physical principles and describe the operation of the thermostat in your home or office.

- C) Text Problem 1.6

1. Note: all page, problem, figure and equation numbers throughout homework assignments 01 to 12 refer to numbering in the 7th edition of Franklin *et al.*.

Text problem 1.6 : Draw a block diagram of the components for an elevator-position control. Indicate how you would measure the position of the elevator car. Consider a combined coarse and fine measurement system. What accuracies do you suggest for each sensor? Your system should be able to correct for the fact that in elevators for tall buildings there is significant cable stretch as a function of cab load.

2 Matlab:

Accessing Matlab :

- UWM has a campus license to Matlab. Students can install Matlab on student-owned devices. To access Matlab see :

<https://uwm.edu/information-technology/software/student>

- To access Matlab via a web browser, see :

<https://www.mathworks.com/products/matlab-online.html>

- Matlab can also be accessed on CEAS CAE Lab computers.

Assignment :

- Open both a web browser and Matlab.
 - Point your web browser to the Control Tutorials for Matlab and Simulink

<http://ctms.engin.umich.edu>

- Navigating the Control Tutorials for Matlab and Simulink

Someone with a penchant for cluttered web pages laid out the first page. But the material is all there. It may be helpful to turn off “Tips” and “Effects” with the check boxes in the upper right.

The web site consists of:

- A basic Matlab tutorial (included in this homework assignment). This is under “Basics -> Matlab” in the upper right.
- Several systems (Cruise Control, Motor Speed, Motor Position, etc.), with materials on modeling and control. T
- Read and carry out the exercises in the section "Matlab Basics." Do not turn in the on-line exercises (the solutions are all given with the text.)

Note: At some points the tutorial directs the student to: "Create an m-file to run and enter the following lines of code: " We will be building and using Matlab skills through out the semester, including using m-files. For now, it is sufficient to copy the lines from the Web Browser, and past them into the Matlab command window.

- A) Turn in the following exercises. With each result that you turn in, include the Matlab code that generated that result. Create the vectors

```
a = [ 1 2 4 2 1]
b = [ 4 3 -1 0 1]
```

And find:

- | | |
|----------------|-----------------|
| A.1) $a + b$ | A.5) $(a == b)$ |
| A.2) $a - b$ | A.6) $\sin(a)$ |
| A.3) $\min(a)$ | A.7) $\exp(b)$ |
| A.4) $(a > b)$ | |

Note: for trig functions, Matlab works in radians.

- B) Create the vector and matrices

```
a = [ 1 2 3 ;
      3 2 3 ;
      0 1 0 ]
b = [ 0 0 1 ;
      0 1 1 ;
      1 1 0 ]
c = [ 1
      2
      3 ];
```

Find:

- B.1) $a + b$
- B.2) $d = \text{inv}(b)$
- B.3) $b * d$
- B.4) $\sin(a)$
- B.5) $a * b$
- B.6) $a .* b$
- B.7) Form the transpose of a
- B.8) What is $a + a$ transpose
- C) What special property does $b * d$ have ?
- D) What is the operation given by $a * b$?
- And what is the operation given by $a .* b$?

E) $a = [2 \ 3 \ 1]$

E.1) What is the result of `roots(a)` ?

E.2) These are the roots of what polynomial ?

E.3) Verify that the roots come from the stated polynomial.

F) For t on the interval from 0..4 plot $\sin(2\pi t)$

Using:

```
>> help title  
>> help xlabel  
>> help ylabel
```

Add labels and a title to your graph.